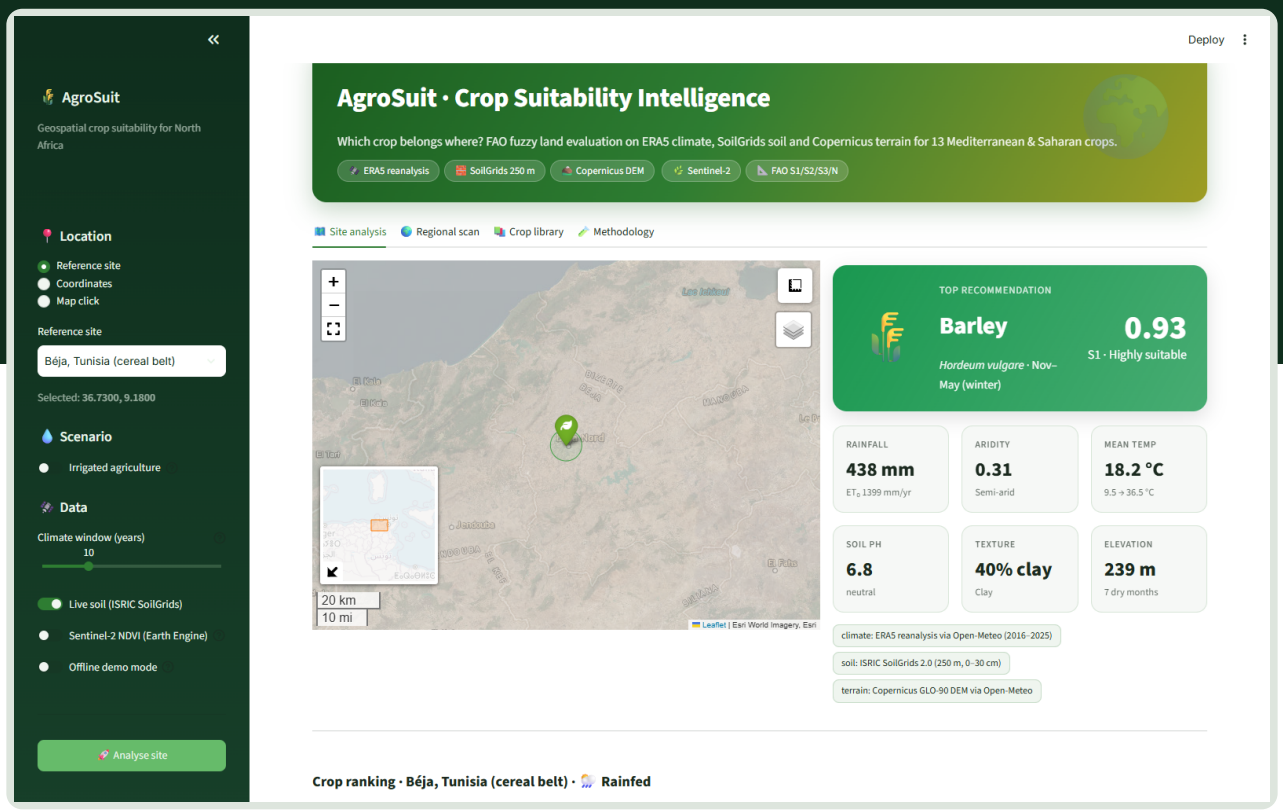
PROJECT REPORT · GEOSPATIAL AI

AgroSuit

Crop Suitability Intelligence for North Africa

FAO fuzzy land evaluation on ERA5 climate, SoilGrids soil and Copernicus terrain for 13 Mediterranean and Saharan crops



Yahya Houimdi
École Nationale d'Ingénieurs de Tunis (ENIT)

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Abstract

Choosing the right crop for a plot of land is one of the highest-impact decisions in North African agriculture, where annual rainfall drops from more than 600 mm in the Tell to less than 100 mm in Saharan oases within a few hundred kilometres. This report presents **AgroSuit**, an open-source geospatial decision-support system that ranks 13 Mediterranean and Saharan crops for any location and explains the result. AgroSuit downloads a ten-year ERA5 climatology including FAO-56 reference evapotranspiration, topsoil properties from ISRIC SoilGrids 2.0 and elevation from the Copernicus GLO-90 DEM, then applies an FAO-style fuzzy land evaluation: trapezoidal membership functions per factor, a weighted geometric mean that follows the law of the minimum, and the FAO classes S1, S2, S3 and N. For every crop the system reports the limiting factor and concrete management advice, and it can compare rainfed and irrigated scenarios and map suitability over whole regions.

The system is delivered as an interactive Streamlit dashboard, a command-line tool, a Jupyter notebook and GIS-ready exports (CSV, GeoJSON, KML, Markdown), backed by a modular Python package with 47 automated tests, Docker packaging and continuous integration. Applied to nine reference sites in Tunisia, Algeria and Morocco, the model reproduces the known agricultural geography of the region: cereals and legumes in the sub-humid north, irrigated citrus in Cap Bon, rainfed olive around Sfax, pistachio in Kasserine and date palm in the Djerid oases.

Keywords: land suitability · fuzzy logic · FAO framework · ERA5 · SoilGrids · GIS · decision support · North Africa

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1. Introduction

1.1 Context

Agriculture in the Maghreb is shaped by a steep climatic gradient and by water scarcity. Over a few hundred kilometres the landscape moves from sub-humid cereal land in north-western Tunisia and the Saïss plain of Morocco, through the olive steppes of the Tunisian Sahel, to hyper-arid oases where agriculture depends entirely on groundwater. Climate change is expected to shift these zones further, which makes it more important to base crop choice on evidence rather than habit.

Planting the wrong crop wastes water, fertiliser and, for tree crops such as olive, almond or pistachio, many years of investment. Yet farmers, extension services and students rarely have a simple tool that combines climate, soil and terrain data for a given point and explains *why* one crop fits better than another.

1.2 Objectives

- **Rank crops** for any coordinate with a 0–1 suitability score and an FAO class.
- **Explain** each result through its limiting factor and the score of every individual factor.
- **Advise** on how to lift a constraint (irrigation deficit, liming, frost and heat management).
- **Compare scenarios**: rainfed vs irrigated, short vs long climate windows.
- **Map** suitability over whole regions to support planning at a larger scale.
- **Engineer** the system to professional standards: modular code, tests, packaging, documentation.

1.3 Scope

AgroSuit covers 13 crops that are representative of North African farming systems (Table 1) and scores them on seven biophysical factors. It targets screening-level decisions, comparable to an FAO land evaluation at reconnaissance scale. It does not replace field soil tests or economic analysis.

Crop	Scientific name	Type	Season	Water need
Durum wheat	*Triticum durum*	Cereal	Nov–Jun (winter)	Medium
Barley	*Hordeum vulgare*	Cereal	Nov–May (winter)	Low
Olive	*Olea europaea*	Tree crop	Perennial	Low
Date palm	*Phoenix dactylifera*	Tree crop	Perennial	High (groundwater)
Citrus	*Citrus sinensis*	Tree crop	Perennial	High
Grapevine	*Vitis vinifera*	Tree crop	Perennial	Medium
Almond	*Prunus dulcis*	Tree crop	Perennial	Low–medium
Pistachio	*Pistacia vera*	Tree crop	Perennial	Low
Fig	*Ficus carica*	Tree crop	Perennial	Low
Tomato	*Solanum lycopersicum*	Vegetable	Mar–Sep (open field)	High
Potato	*Solanum tuberosum*	Vegetable	Two seasons (spring / autumn)	High
Chickpea	*Cicer arietinum*	Legume	Dec–Jun (winter / spring)	Low
Faba bean	*Vicia faba*	Legume	Nov–May (winter)	Medium

Table 1. Crops covered by AgroSuit.

2. Data sources

All inputs are free, global and accessible through public APIs, so the system works anywhere without data preparation. Table 2 summarises the layers.

Layer	Source	Resolution	Variables used
Climate	ERA5 reanalysis (ECMWF), via the Open-Meteo historical API	~25 km, daily	Mean, max and min temperature, precipitation, FAO-56 ET_0
Soil	ISRIC SoilGrids 2.0	250 m, 0–30 cm	pH (H_2O), clay, sand, soil organic carbon
Terrain	Copernicus GLO-90 DEM (via Open-Meteo)	90 m	Elevation
Vegetation (optional)	Sentinel-2 L2A, Google Earth Engine	10–20 m	NDVI, SAVI, NDMI (context only)

Table 2. Data layers and their use.

2.1 Climate: ERA5

ERA5 is the fifth-generation ECMWF reanalysis. It assimilates historical observations into a weather model to give a consistent hourly record of the atmosphere since 1940. AgroSuit requests daily series for *N* full years (default: ten, 2016–2025), computes monthly totals or means per year, then averages across years to obtain a 12-month climatology. Reference evapotranspiration ET_0 is provided directly by Open-Meteo using the FAO-56 Penman-Monteith equation, which allows a proper water balance rather than relying on rainfall alone.

2.2 Soil: SoilGrids 2.0

SoilGrids predicts soil properties worldwide at 250 m with machine learning trained on more than 240,000 soil profiles. Values are stored as scaled integers (for example pH × 10, clay in g/kg) and are converted back to conventional units. AgroSuit computes a thickness-weighted mean of the 0–5, 5–15 and 15–30 cm layers. When the requested pixel is masked (towns, roads, water bodies) the query is retried about 1 km away, and if SoilGrids is unavailable a typical soil for the nearest reference site, or a regional calcareous default, is used. The report always records which source was used.

2.3 Terrain and vegetation

Elevation comes from the Copernicus GLO-90 digital elevation model and is returned with the climate request. In regional scans, cells at or below sea level are treated as sea and removed. Optionally, a Sentinel-2 median composite for the February–May growing season is sampled in a 500 m buffer, using the scene classification layer to mask clouds. The resulting NDVI, SAVI and NDMI describe current vegetation vigour and are shown as context but are not scored, since present land cover does not determine potential suitability.

2.4 Study area

Nine reference sites span the main agro-ecological zones of the Maghreb (Figure 1). They range from semi-arid Meknès (aridity index 0.4) to hyper-arid Tozeur (0.04), where reference evapotranspiration exceeds rainfall by a factor of 24.

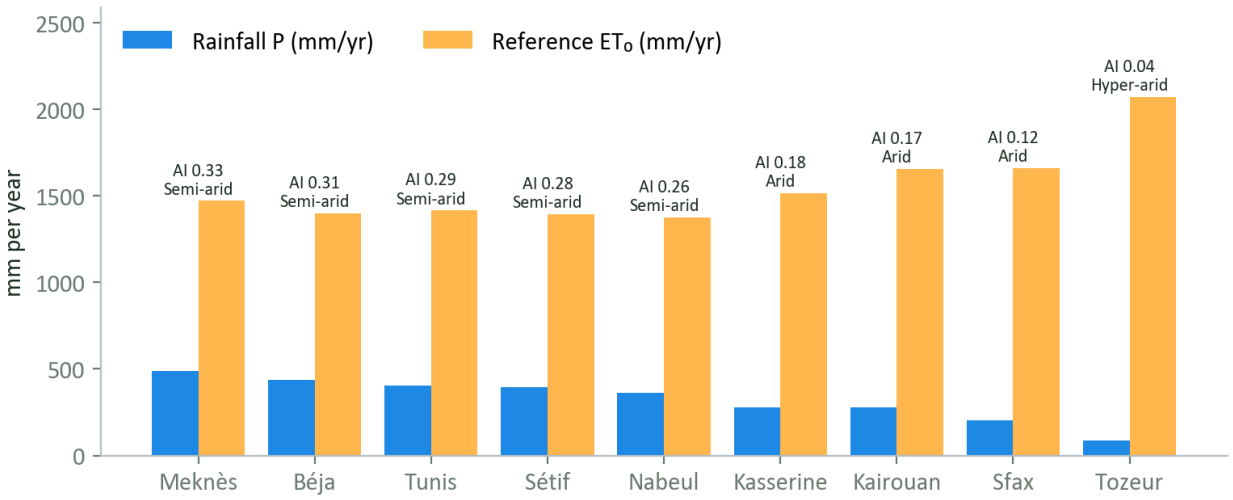


Figure 1. Annual rainfall vs reference evapotranspiration at the nine reference sites (ERA5, 2016–2025), with the UNEP aridity index.

3. Methodology

AgroSuit follows the FAO Framework for Land Evaluation (1976), in which land qualities are matched against crop requirements. Classical implementations use crisp thresholds, so a site with 399 mm of rain is rejected while one with 400 mm is accepted. AgroSuit uses fuzzy set theory instead, giving a gradual and more realistic response. Figure 2 shows the processing chain.

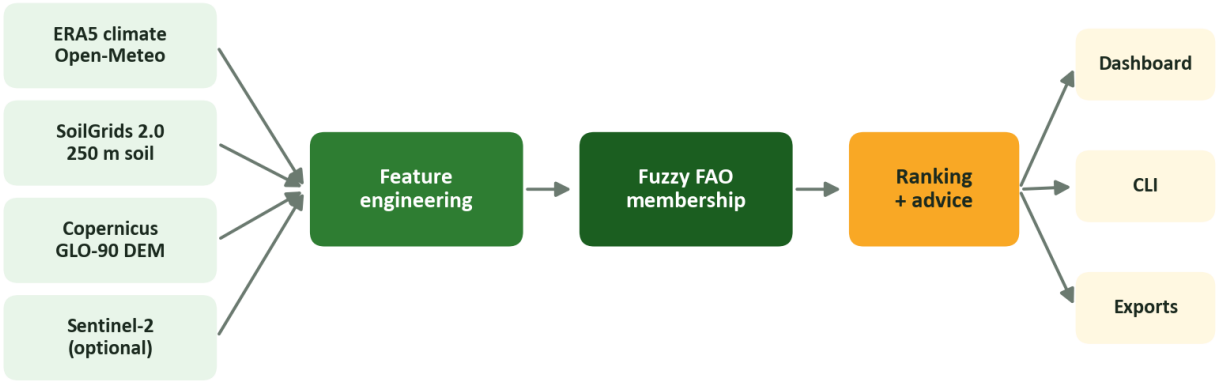


Figure 2. AgroSuit processing pipeline.

3.1 Feature engineering

Indicator	Definition	Role
Annual rainfall P	Sum of the 12 monthly totals	Scored factor
Mean temperature	Mean of the 12 monthly means	Scored factor
Coldest month	Lowest monthly mean temperature	Scored factor (frost, chilling)
Hottest month max	Highest monthly mean of daily maxima	Scored factor (heat stress)
Soil pH, clay	0–30 cm weighted means from SoilGrids	Scored factors
Elevation	Copernicus DEM	Scored factor
Aridity index AI	P / ET_0, UNEP (1997) classes	Context and KPI

Indicator	Definition	Role
Dry months	Months with P < ½ ET_0 (FAO growing-period rule)	Context and KPI
GDD_5	Growing degree days above 5 °C	Context

Table 3. Indicators derived from the raw data.

For offline use, a monthly climatology can be rebuilt from a few climate normals: a cosine annual temperature cycle peaking in late July, a Mediterranean winter-rainfall distribution, and Hargreaves–Samani ET_0 computed from latitude using the FAO-56 extraterrestrial radiation equations:

$$AI = \frac{P}{ET_0} \qquad ET_0^H = 0.0023\, R_a\, (T + 17.8)\, \sqrt{T_{max} - T_{min}}$$

3.2 Fuzzy membership

Each crop defines, for every factor, a trapezoid (*a, b, c, d*): below *a* or above *d* the crop fails, between *b* and *c* it is in its optimum, and in between the membership falls linearly (Figure 3). The ranges are adapted from FAO ECOCROP and regional agronomic literature, and tuned for Mediterranean and semi-arid conditions (see Appendix A).

$$\mu(x) = 0 \;\; (x \le a \text{ or } x \ge d), \quad \frac{x-a}{b-a} \;\; (a < x < b), \quad 1 \;\; (b \le x \le c), \quad \frac{d-x}{d-c} \;\; (c < x < d)$$

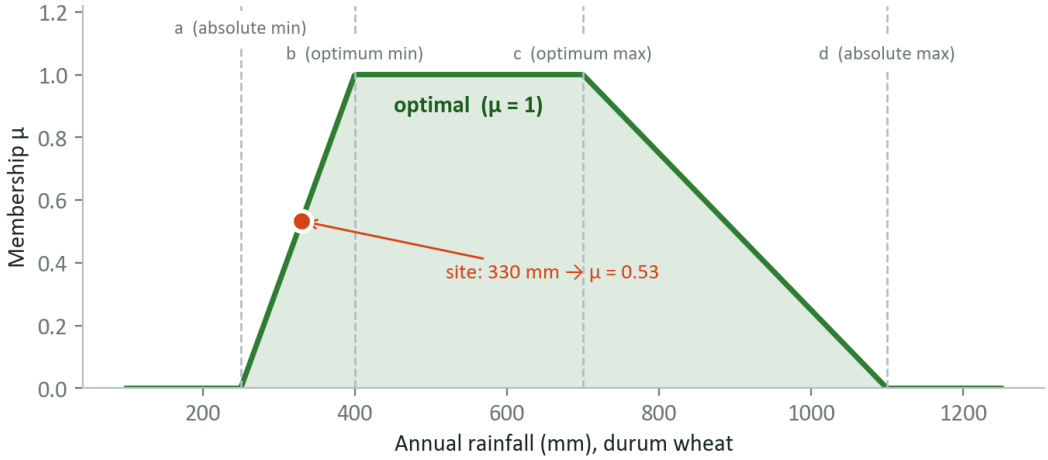


Figure 3. Trapezoidal membership function for durum wheat rainfall. A site with 330 mm scores 0.53.

Under the irrigated scenario, a rainfall *shortage* scores 1 because water is supplied by irrigation, while *excess* rainfall is still penalised, since it favours fungal disease and fruit rot (for example in date palm).

3.3 Aggregation

The factor memberships μ_i are combined into one score *S*:

$$S = \left(\prod_i \mu_i^{\,w_i} \right)^{1/\sum_i w_i} \times \left(0.85 + 0.15\, \bar{C} \right)$$

- The **weighted geometric mean** implements Liebig's law of the minimum: one poor factor drags the whole score down, unlike an arithmetic mean in which a favourable factor can hide a fatal one.
- The **weights** w_i are crop-specific, for example ×1.5 for rainfall in cereals or ×1.4 for the coldest month in frost-sensitive citrus.

- The **centrality term** (C-bar, 1 at the centre of the optimum, 0 at its edges) adds up to 15 % and separates crops that sit comfortably within their optimum from those on its edge, which avoids ties at 1.0.
- If any factor has zero membership, the score is capped at 0.39, so the crop is always classed as not suitable.

3.4 Classification and explanation

Class	Score	Meaning
S1	≥ 0.80	Highly suitable
S2	0.60 – 0.80	Moderately suitable
S3	0.40 – 0.60	Marginally suitable
N	< 0.40	Not suitable

Table 4. FAO suitability classes.

The factor with the lowest membership is reported as the **limiting factor**. Every factor below 0.98 generates a management hint that quantifies the gap and suggests an action, for example:

> **Annual rainfall** 437.9 mm vs optimum 900–1500 mm (score 0.23). Water deficit of ~462 mm/yr: plan supplemental or drip irrigation, or use water-harvesting (tabias, jessour). *Citrus at Béja.*

4. System design and implementation

AgroSuit is written in Python 3.10+ and organised as a package with one responsibility per module. The dashboard, the command-line interface and the notebook are thin front ends over the same two pipeline functions, **analyze_site()** and **scan_region()**.

Module	Responsibility
src/config.py	Paths, API endpoints, FAO classes, regions and reference sites
src/http.py	HTTP client with retries, rate-limit back-off and an on-disk JSON cache
src/weather/open_meteo.py	ERA5 daily series to monthly climatology, single point and batches of points
src/soil/soilgrids.py	SoilGrids query, depth weighting, masked-pixel fallback, USDA texture class
src/satellite/	Spectral indices (NDVI, SAVI, EVI, NDWI, NDMI) and the optional Earth Engine sampler
src/preprocessing/	Climate features, aridity classes, synthetic climatology, FAO-56 radiation, Hargreaves ET_0
src/models/	Crop requirement profiles and the fuzzy suitability engine (scoring, ranking, advice)
src/pipeline.py	analyze_site() returns a SiteReport; scan_region() returns a grid DataFrame
src/visualization/	Plotly charts (ranking, radar, climograph, water balance, curves) and Folium maps
src/reporting/export.py	GeoJSON, KML and Markdown exporters
app/app.py · main.py	Streamlit dashboard · command-line interface

Table 5. Package structure.

4.1 Design principles

- **Separation of concerns:** data access, processing, modelling and presentation are independent and testable on their own.
- **Graceful degradation:** every remote source has a fallback chain (cache, retry, nearby pixel, reference site, regional default), and the report lists the source actually used for each layer together with any warnings.
- **Transparency:** crop requirements are plain data, the model is about a hundred lines of readable code, and every score can be traced to a factor and a range.
- **Reproducibility:** climate windows end at a fixed year, API responses are cached, and all dependencies are pinned in requirements.txt and the Docker image.

4.2 Regional scans

For a region, a regular grid of cell centres is generated and the climate of up to ten points is fetched per request using the multi-location capability of the Open-Meteo API. Sea cells are removed by elevation, every crop is scored in every land cell, and the result is returned as a table that can be mapped or exported as GeoJSON polygons. SoilGrids is not queried per cell because its public API is limited to about five requests per minute, so scans use a regional calcareous default soil. If the climate API rate-limits a batch, the scan continues and reports a partial result.

4.3 Quality assurance

Test file	What is verified
test_suitability.py	Trapezoid maths, FAO thresholds, validity of all crop profiles, S1 and N cases, irrigation logic, missing factors, ranking order, advice text
test_features.py	Spectral indices, extraterrestrial radiation against FAO-56 tables, Hargreaves ET_0, synthetic climatology, aridity classes, leap-year-aware aggregation, texture classes
test_pipeline.py	Offline mode, network-failure fallback, live path with mocked APIs, coordinate validation, grid generation, validity of every export format

Table 6. Automated test suite (47 tests, all passing).

A GitHub Actions workflow runs the suite on Python 3.10 and 3.12, together with an offline smoke test of the CLI, on every push. The dashboard ships with a Dockerfile and a docker-compose configuration.

5. User interfaces

The Streamlit dashboard is the main interface. A sidebar selects the location (reference site, coordinates or map click), the scenario and the data sources. Four tabs present the site analysis, regional scans, the crop library and the methodology.

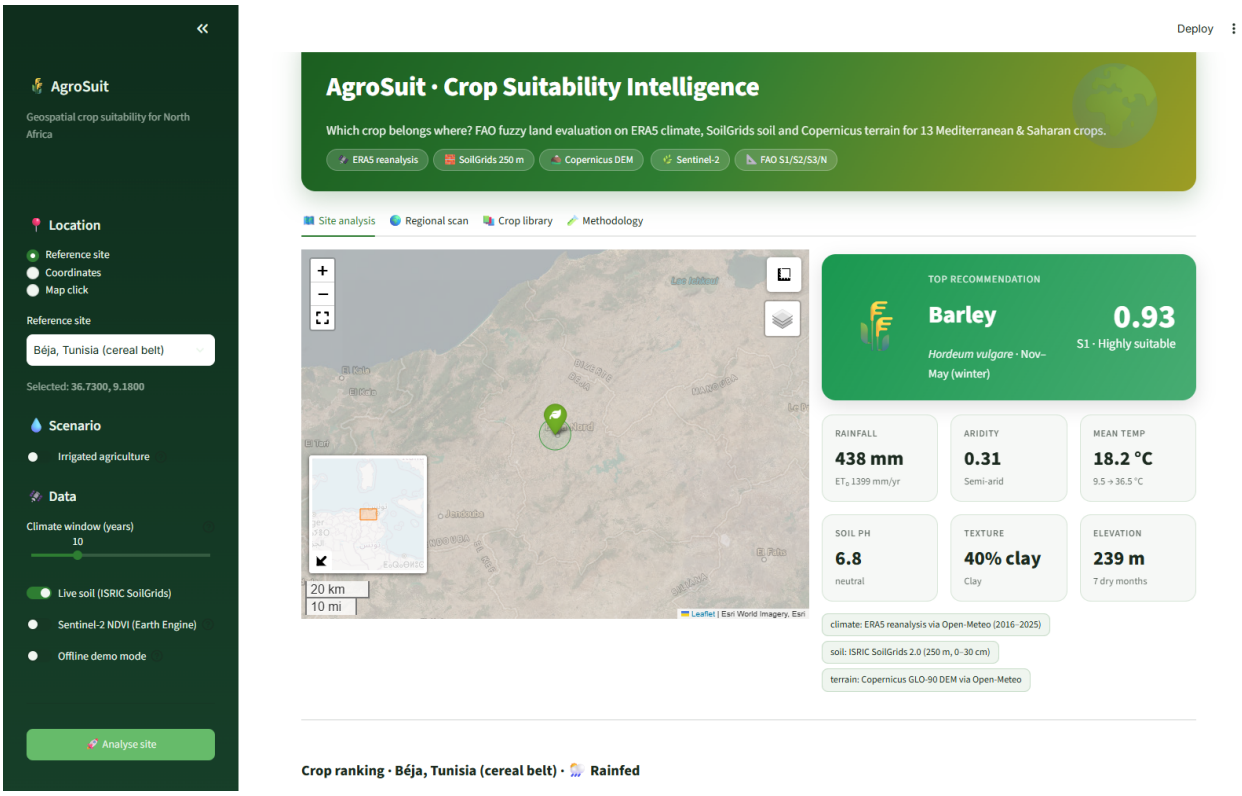


Figure 4. Site analysis for Béja: satellite map, top recommendation, KPI cards and data-source badges.

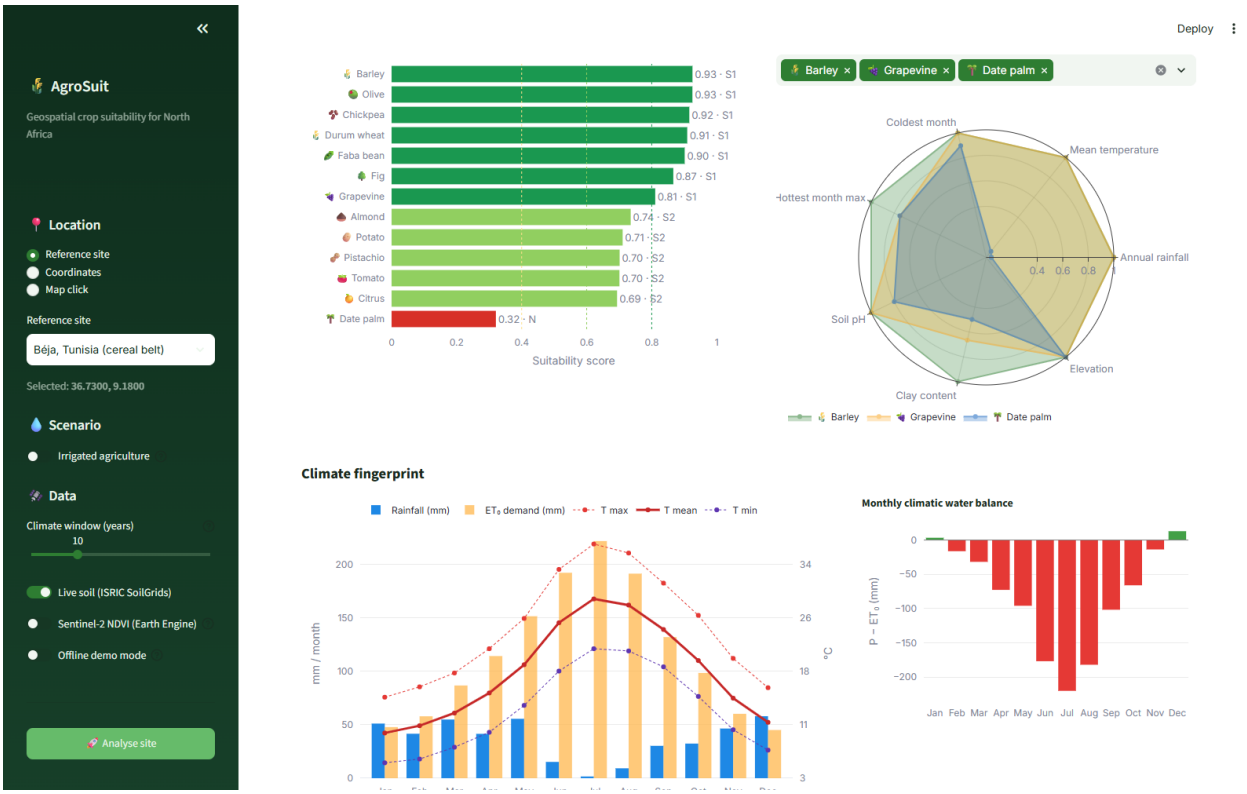


Figure 5. Crop ranking coloured by FAO class, factor radar comparing the best, a middle and the worst crop, and climate fingerprint.



Figure 6. Explanation view: limiting factors with management advice and the site's position on each membership curve.

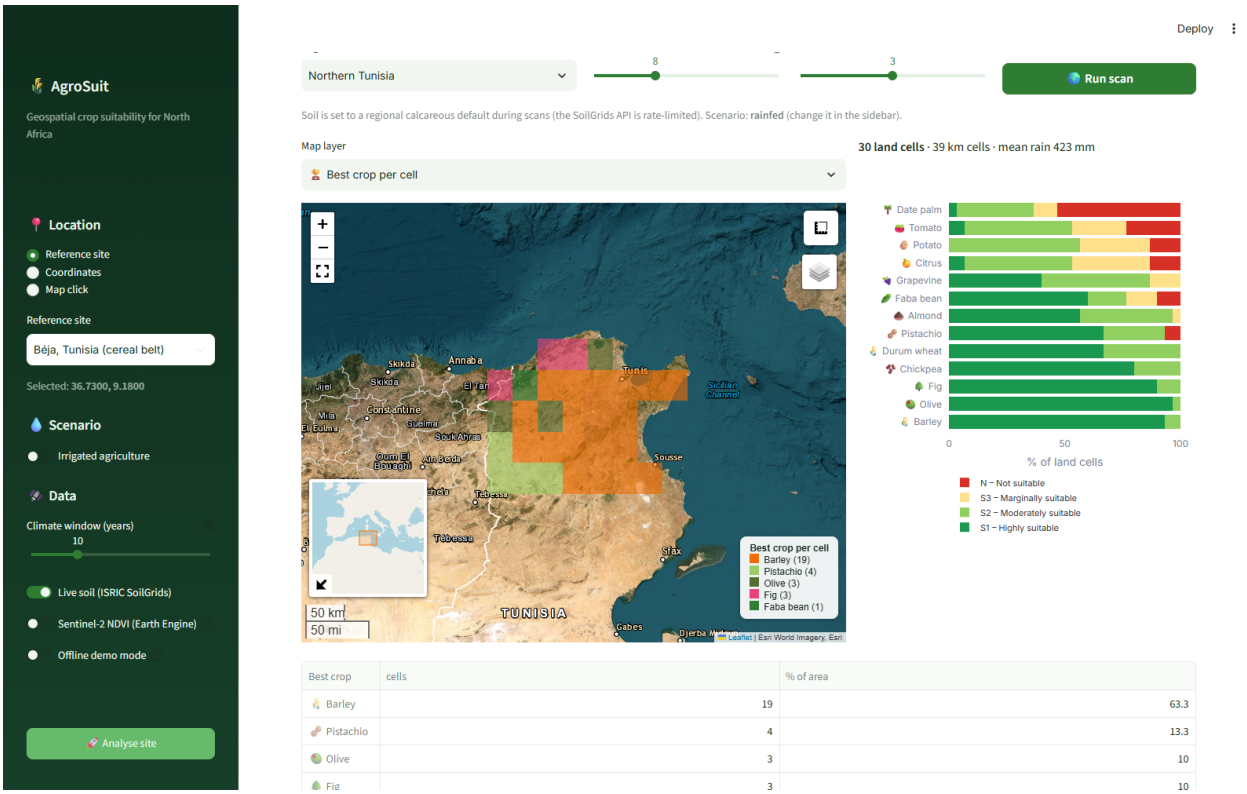


Figure 7. Regional scan of northern Tunisia: best crop per 39 km cell and share of land per FAO class for each crop.

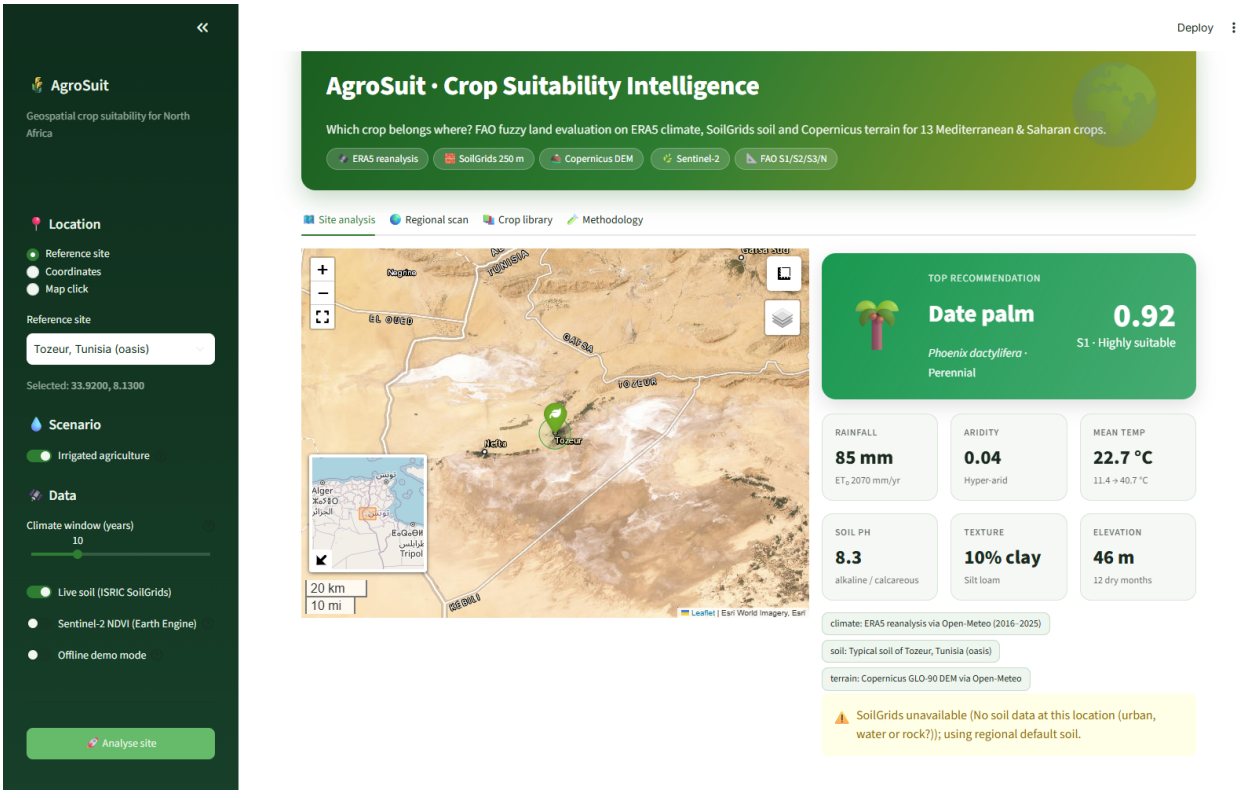


Figure 8. Irrigated scenario at the Tozeur oasis (85 mm of rain, hyper-arid).

The command-line interface offers the same analyses for scripting and batch use, with rich terminal tables, and the **--export** option writes the ranking as CSV, the site as GeoJSON and KML, and a complete Markdown report. Regional scans are saved as CSV and as GeoJSON polygons ready to be styled in QGIS.

6. Results

6.1 Reference sites

Table 7 summarises the analysis of the nine reference sites with live ERA5 climate (2016–2025). Figure 9 shows the full rainfed score matrix.

Site	Rain (mm)	Aridity	Top 3, rainfed	Best, irrigated
Meknès	489	0.33 Semi-arid	Chickpea 0.95, Olive 0.94, Durum wheat 0.94	Chickpea 0.95
Béja	438	0.31 Semi-arid	Barley 0.93, Olive 0.93, Chickpea 0.92	Barley 0.93
Tunis	406	0.29 Semi-arid	Barley 0.92, Olive 0.91, Chickpea 0.91	Barley 0.92
Sétif	395	0.28 Semi-arid	Barley 0.94, Faba bean 0.91, Durum wheat 0.90	Barley 0.94
Nabeul	361	0.26 Semi-arid	Barley 0.91, Olive 0.91, Fig 0.91	Citrus 0.92
Kasserine	277	0.18 Arid	Pistachio 0.96, Barley 0.92, Olive 0.89	Pistachio 0.96
Kairouan	276	0.17 Arid	Barley 0.88, Date palm 0.82, Olive 0.80	Barley 0.89
Sfax	204	0.12 Arid	Olive 0.73, Barley 0.70, Pistachio 0.65	Fig 0.88
Tozeur	85	0.04 Hyper-arid	Date palm 0.92, Pistachio 0.29, Fig 0.28	Date palm 0.92

Table 7. Results at the reference sites.

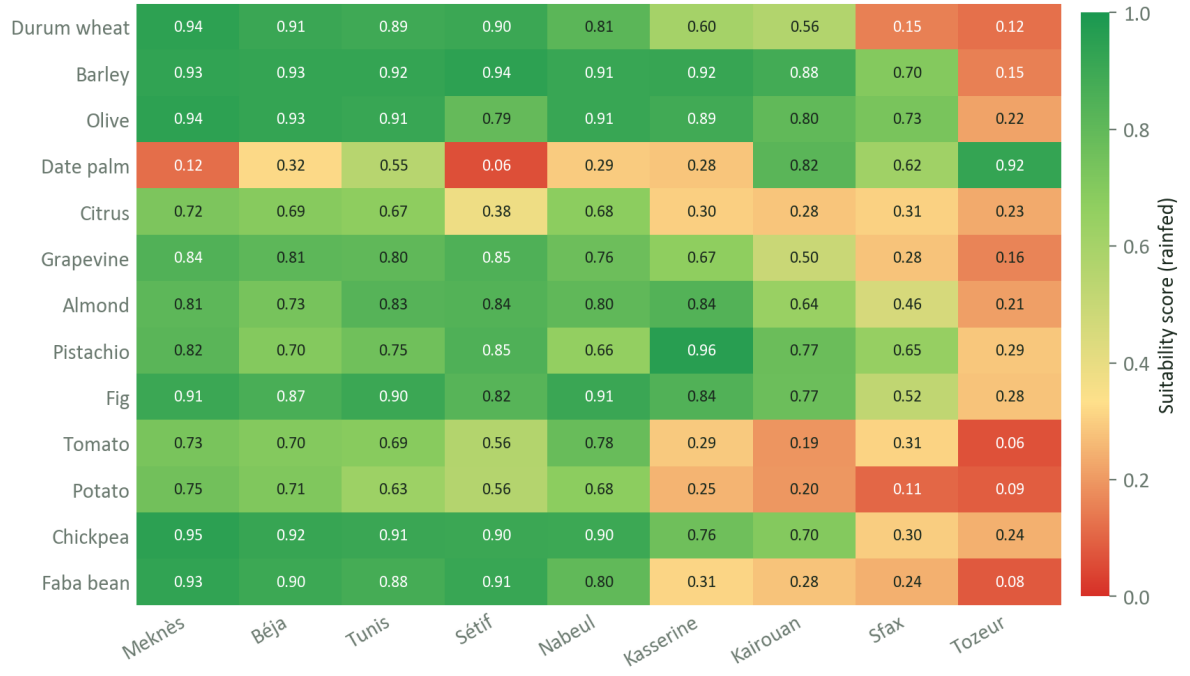


Figure 9. Rainfed suitability of every crop at every reference site, sites ordered from wettest to driest.

6.2 Interpretation

- **Sub-humid and semi-arid north** (Meknès, Béja, Tunis, Sétif): cereals, legumes, olive and fig all reach S1. Chickpea leads at Meknès and barley at Béja, Tunis and Sétif, consistent with the cereal–legume rotations of these regions.

- **Cap Bon** (Nabeul): barley and olive lead under rainfed conditions, but under irrigation **citrus** becomes the best crop (0.92). Cap Bon is indeed Tunisia's main citrus area, where orchards are irrigated.
- **Central Tunisia** (Kasserine): **pistachio** scores 0.96, the highest score in the study, reflecting hot dry summers and cold winters that meet its chilling requirement. The region is a recognised pistachio area.
- **Sahel** (Sfax, 204 mm): **olive** is the best rainfed crop but only reaches S2, limited by water; most annual crops are not suitable. This matches the extensive, widely spaced rainfed olive groves of Sfax.
- **Saharan oasis** (Tozeur, 85 mm): **date palm** is the only suitable crop (0.92), and every other crop fails on rainfall.

6.3 Effect of irrigation

Irrigation removes the water constraint and widens the range of crops in the arid south far more than in the north (Figure 10). At Tozeur the number of crops in S1 or S2 rises from one to seven, at Sfax from four to twelve and at Kairouan from seven to eleven, while the northern sites barely change. The crops that remain unsuitable are those limited by heat or soil rather than water (tomato, potato, faba bean in the oasis), which shows that the model separates water-related constraints from the others.

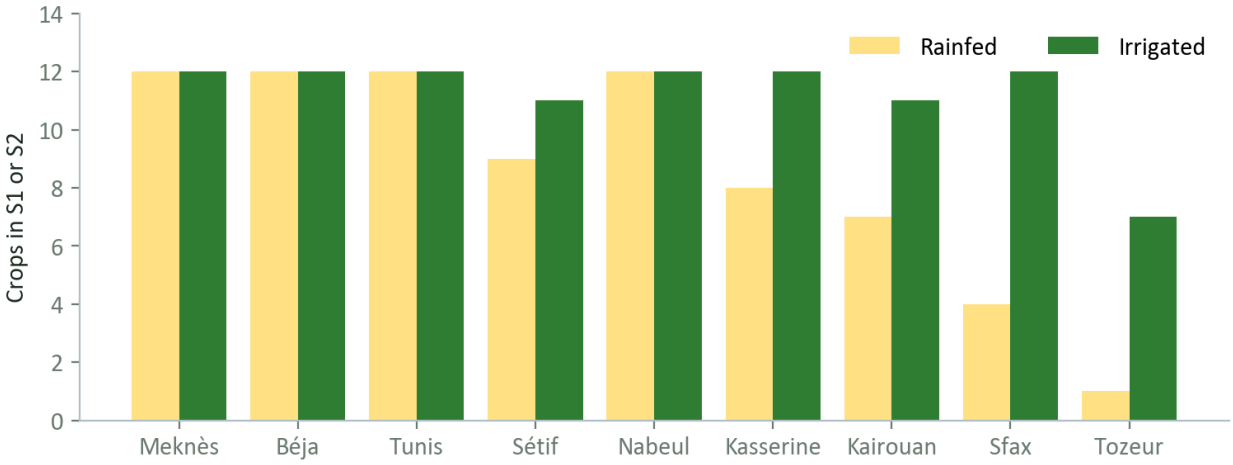


Figure 10. Number of crops classed S1 or S2 at each site, rainfed vs irrigated.

6.4 Regional patterns

In the scan of northern Tunisia (Figure 7, 30 land cells of about 39 km), barley is the best crop in 19 cells, pistachio in the four cells of the drier south-west (around Kasserine and Thala), and olive and fig in a few cells of the north and west. Olive (97 % of cells in S1) and barley (93 %) are the most robust crops, followed by durum wheat (67 %). Date palm is not suitable in half of the cells, and tomato, potato and citrus almost never reach S1 under rainfed conditions.

7. Discussion and limitations

The agreement between model outputs and the actual distribution of crops in Tunisia is encouraging, but it is a plausibility check rather than a formal validation: the crop ranges come from the literature, and a few were adjusted while testing on reference sites. The main limitations are:

- **Generic requirements:** ranges from FAO ECOCROP and the literature are not calibrated against local yield data or cultivar differences (for example early vs late durum wheat).
- **Climate resolution:** ERA5 (~25 km) smooths local climate and tends to underestimate orographic rainfall in the Tell and Rif mountains. Béja, for instance, receives about 600 mm according to station normals, against 438 mm in ERA5.
- **Soil in regional scans:** a default soil is used because of the SoilGrids rate limit.
- **Factors not modelled:** soil salinity, slope, groundwater depth and quality, and socio-economic aspects such as market access and labour.
- **Annual statistics:** seasonal crops are judged on annual indicators. A crop-calendar approach would score only the months of the growing cycle.

8. Future work

- Derive slope and aspect from the DEM and add them as factors.
- Integrate a soil salinity layer with salt tolerance per crop, which is critical in coastal and oasis areas.
- Use SoilGrids WCS rasters to provide spatially explicit soil in regional scans.
- Calibrate requirement ranges against observed land use, from Sentinel-2 crop-type maps or national statistics, possibly with machine-learning tuning of the trapezoids.
- Use CMIP6 climate projections to produce future suitability maps (2050) and identify crops at risk.
- Extend the catalogue to alfalfa, sorghum, onion, pepper and saffron, and add crop-calendar-based seasonal scoring.

9. Conclusion

AgroSuit shows that a transparent, knowledge-based model fed with open global data can produce credible, explainable crop recommendations for North Africa. Combining ERA5, SoilGrids and the Copernicus DEM with FAO fuzzy land evaluation gives scores that are easy to interpret, while the limiting factors and management advice turn the scores into decisions. The project is also a complete software product, with a dashboard, CLI, exports, tests, containerisation and documentation, and a solid base for the calibration and climate-change extensions outlined above.

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Appendix A. Crop requirement profiles

Optimal range (membership = 1) and, in grey, absolute limits (membership = 0) of each factor. Rain: annual rainfall (mm); T: mean temperature; Tcold: coldest-month mean; Thot: hottest-month mean maximum (°C); clay (%); elevation (m).

Crop	Rain	T	Tcold	Thot	pH	Clay	Elev.
Durum wheat	**400–700** 250 – 1100	**13–20** 8 – 25	**2–12** -8 – 17	**25–40** 20 – 47	**6.5–8.2** 5.5 – 8.8	**20–45** 5 – 60	**0–1300** -50 – 2300
Barley	**280–600** 150 – 1000	**12–21** 6 – 27	**0–12** -10 – 18	**25–41** 20 – 48	**6.5–8.4** 5.5 – 9	**12–40** 3 – 60	**0–1600** -50 – 2800
Olive	**300–700** 150 – 1200	**15–21** 12 – 25	**6–12** 2 – 16	**28–37** 22 – 43	**6.5–8.5** 5.5 – 9	**10–40** 3 – 55	**0–800** -50 – 1400
Date palm	**0–150** 0 – 450	**21–30** 18 – 35	**10–18** 5 – 25	**38–46** 32 – 50	**7–8.6** 6 – 9.5	**3–30** 0 – 50	**0–500** -50 – 1000
Citrus	**900–1500** 300 – 2500	**17–25** 13 – 30	**8–15** 3 – 22	**28–36** 24 – 42	**6–7.6** 5 – 8.5	**10–35** 5 – 50	**0–600** -50 – 1200
Grapevine	**400–800** 250 – 1300	**14–20** 10 – 24	**2–10** -5 – 14	**26–35** 22 – 41	**6–8.2** 5.5 – 8.7	**10–35** 3 – 50	**0–1000** -50 – 1600
Almond	**350–700** 200 – 1000	**14–20** 10 – 24	**3–10** -2 – 14	**28–37** 24 – 42	**7–8.4** 6 – 8.8	**8–30** 3 – 45	**0–1100** -50 – 1700
Pistachio	**220–500** 120 – 800	**15–22** 12 – 26	**2–10** -5 – 14	**33–40** 30 – 45	**7.2–8.6** 6.5 – 9.2	**8–30** 3 – 45	**100–1200** -50 – 2000
Fig	**350–800** 200 – 1200	**15–22** 12 – 27	**4–12** -2 – 18	**30–38** 26 – 44	**6.5–8.2** 5.5 – 8.8	**10–35** 3 – 50	**0–900** -50 – 1500
Tomato	**500–900** 300 – 1500	**17–25** 12 – 30	**6–16** 0 – 24	**27–33** 22 – 40	**6–7.6** 5.5 – 8.4	**12–35** 5 – 50	**0–800** -50 – 1500
Potato	**500–800** 300 – 1200	**13–20** 8 – 25	**5–14** -2 – 20	**22–31** 18 – 44	**5.5–7.2** 4.8 – 8.2	**8–28** 3 – 50	**0–1500** -50 – 2500
Chickpea	**350–650** 250 – 1000	**13–22** 8 – 28	**3–12** -5 – 18	**26–38** 22 – 44	**6.5–8.2** 5.5 – 8.8	**15–40** 5 – 55	**0–1200** -50 – 2200
Faba bean	**400–700** 300 – 1100	**12–19** 6 – 25	**3–12** -5 – 16	**24–38** 20 – 44	**6.5–8.2** 5.5 – 8.7	**20–45** 10 – 60	**0–1500** -50 – 2500

Table A1. Trapezoid parameters per crop and factor.

Appendix B. Running the software

Task	Command
Install	pip install -r requirements.txt
Dashboard	streamlit run app/app.py
Analyse a site	python main.py analyze --lat 36.73 --lon 9.18 --export outputs/
Irrigated scenario	python main.py analyze --site Sfax --irrigated
Regional scan	python main.py scan --region "Northern Tunisia" --resolution 10
Tests	pytest -v
Docker	docker compose up --build